

Basic Laws of Boolean Algebra

$A + A = A$	Idempotent law for or
$A * A = A$	Idempotent law for and
$A + B = B + A$	Commutative law for or
$A * B = B * A$	Commutative law for and
$A + (B + C) = (A + B) + C$	Associative law for or
$A * (B * C) = (A * B) * C$	Associative law for and
$A * (B + C) = (A * B) + (A * C)$	Distributive law for and over or
$A + (B * C) = (A + B) * (A + C)$	Distributive law for or over and
$A + 1 = 1$	Law of Union
$A * 0 = 0$	Law of Intersection
$A * (A + B) = A$	Law of Absorption
$A + (A * B) = A$	Law of Absorption
$A * 1 = A$	1 is the identity element for and
$A + 0 = A$	0 is the identity element for or
$\sim(\sim A) = A$	Double negative law
$A + \sim A = 1$	Law of Complement for or
$A * \sim A = 0$	Law of Complement for and
$\sim(A + B) = \sim A * \sim B$	DeMorgan's law
$\sim(A * B) = \sim A + \sim B$	DeMorgan's law